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Indian Sign Language recognition and translation: An Encoder-Decoder Approach

Vibhor Sharma
CSE
National Institute of Technology
Delhi, India
vibhorsharma@nitdelhi.ac.in

Anurag Singh
CSE
National Institute of Technology
Delhi, India
anuragsg@nitdelhi.ac.in

Sabrina Gaito
CSE
University of Milan
Milan, Italy
gaito@di.unimi.it

***Abstract*—The global population of hard-of-hearing individuals is significant and increasing daily, with a major contribution from India. They use different gestures of sign language for communication. It is challenging to understand other sign language if one habitually uses another. The Encoder-Decoder-based architecture generates captions using a Graph neural network for feature extraction. The combination of pose estimation using extracted key points with facial expressions generates a standard way of communication. The novel contribution uses key point extraction and facial expression with context for spatial-temporal graph construction, that is, by using hand movement and other expressions to generate state-of-the-art results in terms of metrics like BLEU, METEOR, and CIDEr.**

***Index Terms*—Sign Language, Indian Sign Language(ISL), Deaf, Dumb, Hard of hearing, LSTM, Deep learning, Graph Convolution Network(GCN)**

I. Introduction

According to the World Health Organization (WHO), approximately 466 million people, or about 6.1% of the world's population, suffer from hearing loss. The number is projected to increase to 630 million by 2030 [9]. Sign language also varies significantly across different countries and regions, each having its vocabulary and grammar [7]. Research has demonstrated that, like spoken languages, sign languages are full-fledged languages [4]. For instance, British Sign Language (BSL) is used in Britain. BSL is a visual language that utilizes both facial expressions and hand movements. It is recognized as an independent language, distinct from English, with its grammatical structure. Additionally, BSL has regional variations across the UK, similar to regional spoken dialects [13]. Indian Sign Language (ISL) is like Hindi and English and can express a language's complexities [14]. ISL fulfills all the criteria of a language [12].

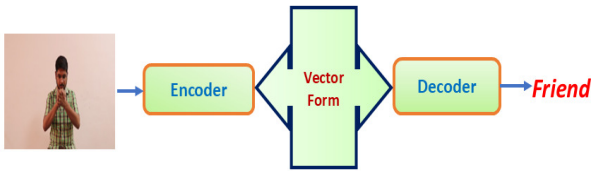

Fig. 1: Encoder-Decoder Model for ISL Translation

II. Related Work

Early methods used template matching to recognize hand shapes and movements by comparing them to a set of predefined templates [1]. Rule-based systems were also developed, where linguistic rules were manually coded to convert recognized signs into textual captions. While these methods provided a foundation for further research, they were limited by their inability to handle variability in signing gesture styles and cultural diversity. [10].
Machine learning models utilized features extracted from video frames and employed classifiers to recognize and classify signs used in ISL. The hand shapes, facial expressions, and other spatial features are extracted using the CNN. To capture temporal dependencies across frames, RNNs, including Long Short-Term Memory (LSTM) networks, have been used [3]. Encoder-Decoder model has effectively generated more accurate translation [6]. An abstract view of the Encoder-Decoder model for ISL translation is shown in Figure 1.

III. Data Set

The publicly available and widely used dataset Indian Sign Language Dataset for Continuous Sign Language Translation and Recognition (ISL-CSLRT Dataset) is used for experiments. The data set contains gloss-level annotations for sign language recognition and translation [8]. The data set comprises video, sentence-level frames, images with annotated words, and spoken language sentences. The signers recorded the videos in a controlled environment. The different signers are shown as in Figure 2. There are 700 videos, 1,036 frames, and 100 spoken language sentences. Seven different signers are used in the Data Set for annotation task. [11].

IV. Proposed System

A. Key Point Extraction using Mediapipe Holistic

MediaPipe helps to identify and locate points of Interest in the video frame. The points of Interest are called vital points. Three ML models are used for Face Mesh, hand tracking, and Pose Estimation for specific tasks for extracting 468 face landmarks, 21 landmarks per hand, and 33 body pose estimations, respectively. The hand-tracking model extracts 21 landmarks for each hand, covering critical joints and points on the fingers and palm. The pose estimation model extracts

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Signer1 Signer 2 Signer 3 Signer4 Signer 5 Signer 6

Fig. 2: Different Signer expresses the same sentences in ISL-CSLRT Data set

33 key landmarks representing the human body's major joints and limbs. The synchronization of all three components gives a holistic understanding of body movements.

B. Feature Extraction

The Encoder-Decoder-based model is trained with the help of extracted features. The movement of hand, body, and face with their position shapes contributes to the spatial features of video frames. The movement of the hand and the trajectory of movement contribute to the temporal features of the object. The trajectory also contributes to the temporal features of video frames.

C. Fusion of key points and features

Hand Pose and Gesture Features are the Key points (landmarks) on the hand, such as the fingertips, knuckles, and wrist, that are tracked to capture hand shapes and movements. Models often analyze the relative position and movement between these points to classify different hand gestures.
Facial Features are Important for recognizing non-manual markers (like facial expressions) that convey grammatical information in sign language. Facial Features include eye gaze, eyebrow movement, and mouth shapes.
Body Pose Features are body landmarks, such as shoulders, elbows, and torso orientation, used to understand body movements and postures integral to some signs.
The most accurate actions can not be detected by only one movement, which combines hand, body, and facial features. The fusion results of the above features are used to train the model. The abstract view of complete process is as in Figure 3.

D. Graph Neural Networks

The graphical data uses Graph Neural Networks. The dependencies between the nodes of a graph can also be captured using the Graph Neural Network (GNN). Graph Neural Networks (GNNs) is a neural network for graphical structured data. GNNs can capture the dependencies between the nodes in a graph.

A Graph Convolutional Network (GCN) layer can be mathematically represented as:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right) \tag{1}$$

where:

- $H^{(l)}$ is called the input feature matrix at the l-th layer.
- $\tilde{A} = A + I$ is the Adjacency matrix of graph with loops, where A is the Adjacency matrix and I is the identity matrix.
- $\tilde{D}$ is known as degree matrix of $\tilde{A}$, defined as $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$.
- $W^{(l)}$ is weight matrix that can be trained at$l$-th layer.
- σ is defined as nonlinear activation function.

The adjacency matrix after normalization is $\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}$. The mathematical expression also ensures that the features of neighboring nodes are scaled appropriately during the aggregation process.

$$h_i^{(k)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} f(h_i^{(k-1)}, h_j^{(k-1)}, e_{ij})\right), \tag{2}$$

where $h_i^{(k)}$ is used to represent the node i at the k-th layer, $\mathcal{N}(i)$ is the set of neighboring nodes i, e_{ij} used to denotes the edge features, and σ is a activation function [5].

V. Graph Neural Networks Approaches

Points are first detected in each frame using a pose estimation model such as MediaPipe or OpenPose. connections. [2].

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right) \tag{3}$$

Where:

- $H^{(l)}$ is called feature matrix that can be input at l-th layer.
- $\tilde{A} = A + I$ is the adjacency matrix with loops.
- $\tilde{D}$ is called degree matrix of the adjacency matrix $\tilde{A}$.
- $W^{(l)}$ is a weight matrix that can be trained at l-th layer.
- σ is an activation function.

Spatio-temporal features are learned through GCNs and are used by the model to understand gestures. Since the gestures change with time, temporal features are correctly captured using temporal features that contribute to accurate sign language recognition and caption generation [15].

VI. RESULTS AND ANALYSIS

The Model was trained for 40 epochs on GPU. The results demonstrate that using GCNs for feature extraction significantly improves the accuracy of sign language recognition as shown in Figure 4. The accuracy for the Model is around 70%, which reflects the correctness of the generated sentence as shown in the Table I

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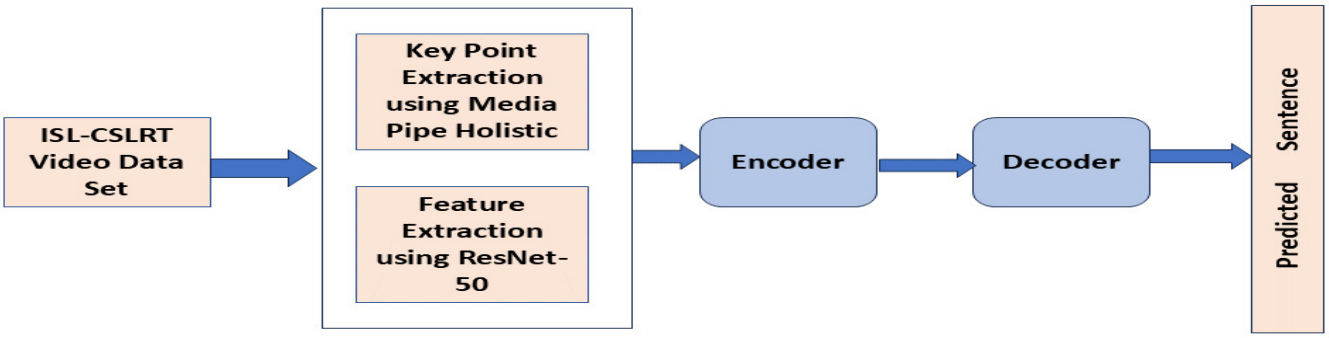

Fig. 3: Block Diagram for the Sign Language recognition and translation.

TABLE I: Comparison of Captions Generated

No.	Ground-truth caption	Translation 1	Translation 2
1.	it does not make any difference to me	it does not make any difference	it does not make any difference.
2.	it was nice chatting with you	It was nice chatting to	it was nice chatting with you.
3.	I was stopped by someone	I was stopped.	stopped by someone.
4.	do you need something.	need something	something.
5.	I am suffering from fever.	I am suffering form ?	I am suffering.

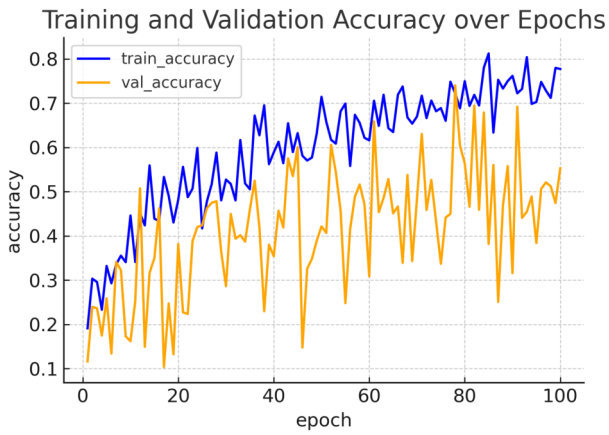

Fig. 4: Validation vs Accuracy

VII. Conclusion

The methodology is critical, enabling those who may not comprehend sign language to access and understand the content. Using Graph Convolutional Networks for feature extraction in Indian Sign Language video captioning provides a powerful method for capturing the spatial and temporal relationships between critical points. In the Future, a Multi-Modal approach that integrates multiple modalities, such as facial expressions and body language, into Sign language translation can be used for a richer interpretation.

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